

# Reviewer Pack — Minimal Norm of Integral Quotients vs DPLL Search Tree Width

Entry 7459ad02d73f · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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## Minimal Norm of Integral Quotients vs DPLL Search Tree Width

Entry ID: 7459ad02d73f **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-22 21:36:10 UTC

### 1. Conjecture

**Field A** (mathematical branch): Integral Representation Theory **Field B** (complexity object): Complexity Theory: DPLL Search Tree Complexity

#### Statement:

[‘For every Boolean formula  $F$  with  $n$  variables, the minimal norm of an integral quotient in the integral representation of  $F$  is  $\Theta(\log(n))$ , where the norm is defined as the maximum absolute value of the coefficients.’, ‘For all instances with property  $P$ , property  $Q$  holds if and only if the DPLL search tree width for that instance is at most  $\alpha \log(n)$ , where  $\alpha$  is a constant independent of  $n$ .’, ‘The statement above is falsified if there exists an instance  $F$  with  $n$  variables such that the minimal norm of its integral quotient is less than  $\Theta(\log(n))$  or the DPLL search tree width is greater than  $\alpha \log(n)$  for some  $\alpha$ .’]

#### Rationale (proposer’s reasoning):

[‘Integral representation theory provides a framework to study algebraic properties of Boolean formulas. The norm of an integral quotient could capture essential algebraic information that influences the structure of the DPLL search tree.’, ‘The DPLL search tree width is a measure of the complexity of solving a Boolean formula. If there were a direct relationship between the norm of integral quotients and the search tree width, it might lead to new insights into the complexity of satisfiability problems.’]

**Taxonomy category:** INTEGRAL\_REPRESENTATION (status at proposal time: )

### 2. Pre-registration (Popper-style)

**Hash (SHA-256 prefix):** db324dac50fb0f66

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

#### Acceptance criterion:

The conjecture is supported if for all instances  $F$  with  $n$  variables ( $n \leq 40$ ), the minimal norm of its integral quotient is within  $\Theta(\log(n))$  and the DPLL search tree width is at most  $\alpha \log(n)$  for some constant  $\alpha$ , where both metrics are computed using 30 different random

seeds. The conjecture is falsified if any instance has a minimal norm outside of  $\Theta(\log(n))$  or if there exists an instance with a DPLL search tree width greater than  $\alpha \log(n)$ .

### 3. Barrier filter (F1)

**Final:** PASS

| Barrier        | Verdict | Confidence | LLM <sub>1</sub> | LLM <sub>2</sub> |
|----------------|---------|------------|------------------|------------------|
| RELATIVIZATION | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |
| NATURAL_PROOFS | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |
| ALGEBRIZATION  | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |
| KARP_LIPTON    | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |

### 4. Novelty audit

**Verdict:** NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

**Search queries** (3): - "integral representation theory" AND "DPLL search tree complexity"  
- "minimal norm of integral quotients" AND " $\Theta(\log(n))$ " - "property P and Q" AND DPLL AND search tree width AND  $\alpha \log(n)$

### 5. Empirical test harness

**Multi-seed protocol:** 5 seeds (11, 23, 37, 53, 71)

**Sandbox:** isolated subprocess, stdlib-only,  $\leq 90$ s wall time per cycle **Execution:** rc=1, elapsed=0.1s

#### 5.1 Generated Python source

```
import random
import math

def generate_formula(n):
    variables = [f'x{i}' for i in range(1, n+1)]
    clauses = []
    for _ in range(n):
        clause = random.sample(variables + [-v for v in variables], 2)
        clauses.append(' '.join(clause))
    return ' & '.join(clauses)

def integral_representation(formula):
    norm = 0
    for clause in formula.split(' & '):
        clause_norm = sum(abs(int(c[1:])) if c[0] != '-' else -int(c[1:])) for c in clause.split()
        norm = max(norm, clause_norm)
    return norm

def dpll_search_tree_width(formula):
    # Simplified DPLL search tree width calculation
    return len(formula.split(' & '))

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.randint(5, 40)
    formula = generate_formula(n)
```

```

norm = integral_representation(formula)
width = dp11_search_tree_width(formula)

metric_name = "Minimal Norm of Integral Quotients"
metric_value = norm
instances_tested = 1
conjecture_holds = norm <= math.log(n) and width <= 2 * math.log(n)
counterexample = "" if conjecture_holds else f"Formula: {formula}, Norm: {norm}, Width: {width}"

return {
    "metric_name": metric_name,
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2**i - 1 for i in range(5, 8)]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(r['metric_value'] for r in results) / len(results)
    std_value = math.sqrt(sum((r['metric_value'] - mean_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r['conjecture_holds']) / len(results)

    if all(r['conjecture_holds'] for r in results):
        result = f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}"
    elif support_fraction >= 0.8:
        result = f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}"
    else:
        first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])
        result = f"RESULT: FALSIFIED counterexample=\"Formula: {results[0]['counterexample']}\" f"

    print(result)

```

## 6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

## 7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp\_sandbox/test\_4bad1920.py", line 65, in <module>

```

    trial_result = run_trial(seed)
    ~~~~~

```

File "/home/ludo/Scrivania/SEC/research/pvsnp\_sandbox/test\_4bad1920.py", line 40, in run\_trial

```

    formula = generate_formula(n)

```

^^^^^^^^^^^^^^^^^^^^

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4bad1920.py", line 22, in generate_form
    clause = random.sample(variables + [-v for v in variables], 2)
    ^^^
```

TypeError: bad operand type for unary -: 'str'

## 8. Critic adversarial review

**Critic verdict:** CHALLENGE

**Critic reasoning:**

skipped: test crashed before producing data

## 9. Final verdict & safety rail

**Verdict:** INCONCLUSIVE

**Reasoning:**

The test code crashed before producing data, which prevents us from verifying the conjecture's conditions. | next: Investigate and fix the error in the test code to ensure it can complete without crashing.

## 11. Audit log (LLM calls)

**Total LLM calls:** 9

| # | Phase           | Provider      | Model            | Tokens in | out | Latency (ms) |
|---|-----------------|---------------|------------------|-----------|-----|--------------|
| 1 | propose         | ollama_remote | glm4:latest      | 0         | 0   | 14863        |
| 2 | preregistration | ollama_remote | glm4:latest      | 0         | 0   | 9870         |
| 3 | novelty         | ollama_remote | glm4:latest      | 0         | 0   | 8500         |
| 4 | novelty         | ollama_remote | glm4:latest      | 0         | 0   | 8979         |
| 5 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 14676        |
| 6 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 7938         |
| 7 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 7865         |
| 8 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 8039         |
| 9 | judge           | ollama_remote | glm4:latest      | 0         | 0   | 27091        |

**Totals:** 0 input tokens, 0 output tokens, ~\$0.0000 cost, 107819 ms total latency. Provider mix: {'ollama\_remote': 9}

(full prompt+response transcripts available in `research/audit/7459ad02d73f.jsonl`)

## 12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/7459ad02d73f.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

**Sandbox file:** `research/pvsnp_sandbox/test_*.py` (per-cycle)

**Replay tarball:** `research/replay/7459ad02d73f.tar.gz` (if generated)